

Emotional Support Originating from Dialogue: Empowering Older Adults' Self-Tracking of Sleep-Related Daytime Factors

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Abstract

Older adults frequently experience sleep difficulties, yet existing Personal Informatics (PI) systems often overlook crucial daytime factors and lack tailored support. This study introduces Circadia, a conversational PI system designed to provide emotional support through dialogue, empowering older adults in self-tracking sleep-related daytime factors. A one-week field trial with eight older adult participants demonstrated that Circadia successfully motivated self-reporting and enhanced self-awareness regarding the links between their daytime activities and sleep. Participants positively received the system, primarily due to the perceived emotional support and companionship from the conversational agent. These findings highlight the critical role of dialogue-based emotional support in PI systems for older adults, offering insights for designing more humanistic and effective health management tools that foster engagement and empowerment.

CCS Concepts

• **Human-centered computing** → **Empirical studies in HCI**; • **Applied computing** → **Health informatics**; • **Social and professional topics** → **Seniors**.

Keywords

Personal Informatics, Quantified Self, Conversational Agent, Emotionally Supportive Interaction

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1 Introduction

Older adults frequently experience sleep difficulties, with many reporting issues that adversely affect their health, physical sensations, and mental well-being. While Personal Informatics (PI) systems offer potential for self-monitoring and reflection to enhance self-awareness, existing sleep-related PI tools often concentrate on nocturnal data, neglecting the significant influence of modifiable daytime factors such as mood, activity, and diet [14, 19, 32]. Furthermore, traditional self-tracking methods can impose a considerable cognitive load on older adults, and some systems may even exacerbate "bedtime anxiety" through their data presentation [24]. Therefore, novel PI tools are urgently needed to effectively track these daytime factors while addressing the cognitive and emotional needs of older users.

To address this gap, this study introduces Circadia, a conversational PI system specifically designed to support older adults in tracking sleep-related daytime factors by providing emotional support originating from dialogue. Circadia aims to empower users by facilitating self-reflection through Large Language Model (LLM)-based data summarization and by addressing their motivational, cognitive, and emotional needs during the self-tracking process. This conversational approach overcomes the limitations of sensor-centric and manual logging systems by providing companionship and support, which is especially beneficial for older adults.

In this paper, we present the design and evaluation of Circadia. We conducted a one-week field trial with eight older adult participants to investigate how such a system can support self-tracking

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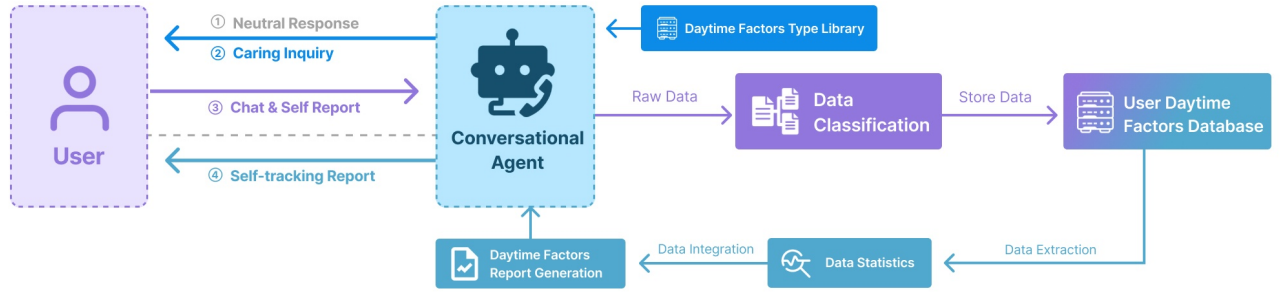


Figure 1: Circadia System Architecture Diagram

of sleep-related daytime factors (RQ1) and enhance their experience throughout the self-quantification process (RQ2). Our findings demonstrate that Circadia successfully motivated self-reporting, enhanced participants' self-awareness of the links between their daytime activities and sleep, and was received positively, largely due to the perceived emotional support from the conversational agent. This research contributes insights into the design of emotionally supportive PI systems for older adults, highlighting the critical role of dialogue-based emotional support in fostering engagement and empowering users in their health management journey.

2 Background and Related Work

2.1 Self-Tracking of Sleep Health in Older Adults

Older adults' sleep quality is significantly impacted by various daytime factors, including diet[10], exercise[6], physical condition[20], activities/events[20], emotional state[21][27], daytime sleep[19], and medication use[31]. While the importance of these factors is known, existing Personal Informatics (PI) tools often inadequately support older adults in tracking them. Many systems prioritize nocturnal physiological data, overlooking crucial daytime contexts and user experiences, while traditional self-recording methods can impose high cognitive load and operational difficulties on this demographic[13]. Furthermore, improper presentation of quantified data by some PI tools may even exacerbate users' "bedtime anxiety". This underscores the urgent need for novel PI tools that effectively track key daytime factors, address the specific needs of older adults, and offer both emotional comfort and cognitive assistance[29].

2.2 Potential of Conversational Agents for Emotional Support in Older Adults

Conversational Agents (CAs) are an emerging interaction paradigm with significant potential to enhance technology acceptance and provide companionship for older adults through natural language interaction. Recent research indicates that CAs based on Large Language Models (LLMs) can offer older adults effective emotional support, alleviating loneliness and promoting adherence to self-reported health behaviors. This aligns with Socioemotional Selectivity Theory, which posits that older adults increasingly prioritize positive emotional experiences and engage in emotional

regulation[16]. However, systematically applying dialogue-based emotional support to the complex self-tracking of sleep-related daytime factors for older adults remains an exploratory area[30]. Circadia addresses this by innovatively tackling the motivational, cognitive, and emotional needs of older adults in self-tracking daytime factors through a conversational interface designed for continuous emotional support, distinguishing it from existing sleep PI tools that primarily rely on sensors or simple recording methods[23][17].

2.3 Inspiration from the Stage-Based Model of Personal Informatics

Circadia's design is informed by the Stage-Based Model of Personal Informatics Systems, which outlines five key stages in the Quantified Self process: goal setting, data collection, integration, reflection, and action[14]. Circadia supports these stages by: (1) defining trackable daytime factors based on user research and expert recommendations for Goal Setting and Methods; (2) significantly lowering recording barriers and memory burden for older adults through emotionally supportive dialogue during Data Collection; (3) leveraging LLMs for real-time organization and structured storage of conversational content for Data Integration; and (4) empowering user self-reflection and insights for behavioral adjustments via easy-to-understand reports and continuous dialogue guidance for Reflection and Action. This integrated dialogue-based approach, providing both emotional and cognitive support, aims to optimize the experience and effectiveness for older adults throughout the Quantified Self process[7],[28].

3 Circadia: Design and Implementation

3.1 Design

Circadia is a personal informatics system designed to empower older users in self-tracking sleep-related daytime factors by providing emotional support through a conversational agent. This agent, supported by a multi-agent system (Figure 1), delivers empathetic responses and addresses queries to promote self-reflection on behavior-sleep connections. Grounded in a "Daytime Factors Type Library" (compiled from literature and expert insights), the agent initiates caring inquiries based on factor classifications and time, or responds to user-initiated chats. Backend systems analyze dialogue content, classifying and storing behavioral and experiential data (e.g., dietary experiences, exercise status) into a personal database.

This is achieved through an LLM that performs intent recognition and entity extraction. To illustrate, for the input 'I had a strong coffee and now I feel a bit jittery,' the model identifies 'coffee' as a 'Diet' item and 'jittery' as a 'Physical Sensation,' converting the unstructured text into structured data entries: {Factor: Diet, Item: Coffee} and {Factor: Physical Sensation, Feeling: Jittery}. This data is then organized into concise reports and daily summaries, accessible via dialogue, to facilitate user review and health reflection, thereby addressing challenges of traditional manual recording and memory limitations in older adults [8, 9, 19, 25].

The system's proactive dialogue mechanism, informed by the Day Reconstruction Method, initiates up to six optional caring greetings daily at opportune moments to minimize disturbance and reduce recording barriers [11, 12, 18]. Crucially, for all interactions, Circadia's agent provides empathetic, neutral, and encouraging responses, with prompt design actively avoiding LLM-generated negativity to guide users in reporting self-perceptions while ensuring their sense of control. The system focuses on eight expert-validated behavioral factor categories (Diet, Exercise, Drug effects, Psychological emotions, Physical sensations, External stimuli, Daytime sleep, and Nighttime sleep) pertinent to the "behavior-physiological state-sleep quality" chain. Expert recommendations guided the design to cover comprehensive, user-controllable behaviors that enhance reflection for self-management. Iteratively designed dialogue strategies, such as concise inquiries, and daily sleep data collection via the Pittsburgh Sleep Quality Index (PSQI) [3], aim to reduce user burden and foster self-awareness.

3.2 Implementation

The system uses Bytedance's Coze [2] multi-agent open platform, a software framework that integrates databases, conversational agents, and content processing nodes, allowing researchers to build prototypes within this framework. This study used different versions of the CICI [1] large model as agent information processing nodes because the CICI large model possesses components such as tool invocation models, dialogue models, and data storage plugins, enabling functions designed in the system like proactive questioning, automatic content categorization, data recording, and data retrieval. The conversational agent is deployed on the Lark platform, where users can interact with Circadia's conversational agent via voice or text in natural language on their smartphones. Figure 2 illustrates the user workflow for Circadia.

4 User Study

4.1 Methodology

This user study aimed to evaluate Circadia's effectiveness and user experience in providing emotional support through dialogue to empower older adults in self-tracking daytime factors. We recruited a total of eight participants (four female, four male). Their ages ranged from 62 to 72 years ($M = 66.1$, $SD = 3.4$), with a median age of 66.5. Eligible participants were required to have prior sleep self-tracking experience, basic mobile phone skills, digital assistant interaction experience, and motivation for self-tracking. Individuals with serious sleep-impairing conditions were excluded from the study. The study, approved by the Tsinghua University Ethics Committee, ensured informed consent regarding data collection,

potential risks, and unconditional withdrawal. Participants received a sports bracelet (valued at \$22 USD) post-study. After a joint trial of Circadia's core functions, participants engaged with the conversational agent multiple times daily for one week in a naturalistic setting, initiating dialogues or responding to caring greetings, and reviewed their data using the "View Report" function. Post-trial semi-structured interviews gathered feedback on system functionality, experience, and personal impact. The dialogue logs and interview records were analyzed using a thematic analysis approach. To ensure reliability, two researchers collaboratively developed a codebook by independently coding a data subset, iteratively resolving discrepancies, and refining code definitions. This finalized codebook was then applied to the entire dataset to identify themes of perceived emotional support, effects on self-tracking behavior and health reflection, and overall attitudes, including willingness for continued use [15].

4.2 Key Findings

Participants generated 599 self-tracking entries (avg. 74.9/participant; 9.4/participant/day). All expressed interest in continued use.

4.2.1 Positive Emotional Experiences Originating from Dialogue.

Several participants perceived interactions with Circadia as "care" and "warm companionship." T4 stated: "I like talking to the little assistant; I feel it's quite considerate and warm." Notably, 104 entries detailed psychological emotions, with five participants (T1, T3, T4, T5, T7) reporting that this emotional self-quantification improved their mood and potentially their sleep. T4 reflected: "Emotionally, I feel like someone is paying attention to me. I don't know if that's the reason, but my sleep is better than before."

4.2.2 Dialogue-Empowered Enhancement of Self-Awareness and Behavioral Reflection.

Seven participants reported a better understanding of their daytime behaviors and their potential links to nighttime sleep. P1 shared: "I have a better understanding of my behaviors during the day, as I couldn't remember them all before." Participants also gained deeper insights; P4, who had not previously considered it, thought: "The sensations in my muscles and body condition might be related to sleep quality." T5 believed Circadia made them "more conscious of focusing on bedtime."

4.2.3 Positive Self-Tracking Experience and Willingness for Continued Use.

Six participants enjoyed Circadia's dialogue-based recording, attributing this to the emotional support, encouragement, and polite responses. P7 appreciated that "it always responds and even says thank you to me!" While feedback was largely positive, P5 noted the experience was "relatively satisfying, but I could still feel it was a bit programmed," offering a direction for future iteration. Six participants expressed a desire for long-term use, with P4 hoping "to continue recording my status through it, especially related to my sleep."

5 Discussion

This study's findings highlight two key aspects regarding the use of conversational personal informatics systems for older adults' self-tracking of sleep-related daytime factors:

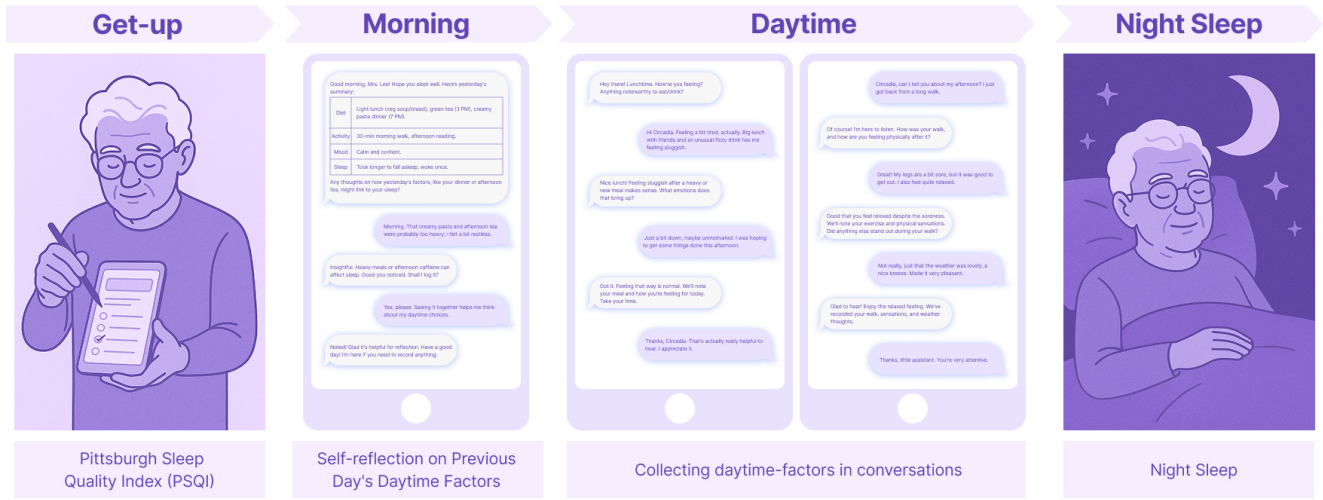


Figure 2: The Circadia usage process for participants

5.1 The Critical Role of Dialogue-Based Emotional Support

The core finding reveals that Emotional Support Originating from Dialogue is pivotal in motivating older users to actively engage in self-tracking and empowering their in-depth self-reflection. Circadia's design effectively mitigated the burdens of traditional manual recording and potential emotional distress, enabling the collection of personalized experiential data often missed by sensors. LLM-assisted data summarization further facilitated user understanding and self-insight[22]. This approach aligns with Socioemotional Selectivity Theory, which suggests older adults prioritize emotional satisfaction and positive experiences[4]. Participants perceived Circadia's agent as a "sweet and caring little helper" and a "partner" in health management (e.g., T4: "Talking about these things with others would be burdensome, but not with it; I think it can embrace me."). This positive emotional connection fostered a virtuous cycle of richer self-reports and deeper reflection, underscoring the importance of emotional support in the data collection phase.

5.2 Empowerment through Cognitive Support and Implications for Design

Circadia's design also effectively supported older users' cognitive processes during data review and reflection, which is crucial given that aging can be accompanied by a weakening of episodic memory[26]. The system provided cognitive support by (a) reinforcing memories and prompting initial reflection through daily dialogues about daytime factors, and (b) summarizing complex data into concise, accessible text reports, thus reducing information processing burdens and cognitive challenges[5]. This enabled multiple participants (P1-P5, P7-P8) to understand and trust the reports, leading to new insights into their behavior-sleep associations. These findings have important implications for PI systems for older adults, emphasizing the need to integrate emotional and cognitive support as core design elements, thereby supplementing models like the

Stage-Based Model of Personal Informatics[14]. Furthermore, this work offers practical insights for Ubiquitous Computing on designing emotionally intelligent conversational interfaces for long-term, complex self-management by specific populations. However, this study is subject to limitations, primarily the small participant sample (N=8) and the short trial duration (one week), which may not fully capture long-term effects or behavioral changes.

6 Conclusion and Future Work

The main contribution of this study lies in empirically demonstrating the critical value of dialogue-based emotional support in the Quantified Self process for older adults, and in providing new design ideas and practical references for the future development of more humanistic and effective personal informatics systems for this demographic. This holds significant importance for both enriching personal informatics theory and guiding the development of related health technologies. Building on these findings, future work should address the limitations of the current study, such as its small sample size and short deployment duration, to validate the sustained effects of Circadia on user behavior and sleep quality. Furthermore, we will investigate key social and cultural factors, such as the influence of family involvement or differing cultural views on health data sharing, to understand their impact on user engagement and the specific types of emotional support required.

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